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Abstract

The fast development of digital technologies has encouraged organizations to resort more to security technologies to protect information and systems and infrastructures. Security technologies include network security, endpoint protection, data encryption, cloud security, and the new tools in this category, including AI-driven solutions and blockchain-based security mechanisms. The report will set out to examine the importance of these technologies in critical mitigation, organizational resilience, and cyber incident prevention. Two case studies are performed in the present day, the 2023 ransomware of Royal Mail in the United Kingdom and the 2024 cloud breach of Microsoft Azure (Storm-0558 follow-up). Through such cases, it is seen how vulnerable both traditional and cloud-based infrastructures are and why organizations must implement an all-encompassing security technologies. The report has included comprehensive literature review, identification of technical, organizational, and legal obstacles and best practices that can be applied. The results imply that companies should employ a multi-layer security methodology which entails technological, human, and procedural security. Experiences of recent breaches show that in order to protect sensitive data and preserve the continuity of operations, constant monitoring, the analysis of risks and the active implementation of new technologies are needed. The report ends with a recommendation to organizations on how they can improve their security posture in a more complicated threat environment.

Acknowledgement

I would like to take this opportunity to acknowledge heartily to all of you who helped me in the completion of this coursework on **Security Technologies**. To begin with, I would like to express my deepest gratitude to my module leader **Mr. Anish Pudasaini**, who has helped me immensely and provided me with helpful feedback and constant motivation to continue my study. His lectures, tips, and work guidance have assisted me to come up with a clear idea of the topic and what was needed in this report.

I also would like to recognize the learning resources, the digital libraries, an excellent academic environment that was offered at Islington College and London Metropolitan University as the prerequisites to score a successful research. I am grateful to the authors, researchers, and organizations, the publications of which, journals, and online resources were used as essential contributors to the literature review and analysis of the case study.

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1. Introduction of Security Technologies

As the use of digital systems grows and organizations are becoming more dependent on them, a more threatening of environment has emerged with ransomware, phishing, insider threats, and cloud misconfigurations to emerging. Security technologies are important tools and solutions meant to identify, avert and react to cyber threats to guarantee confidentiality, integrity and availability of the information assets. The building of a strong security posture should include network security, endpoint protection, encryption, identity and access management, cloud security and monitoring systems (Valorem Reply, 2025).



Figure 1 : Evolution of Security Technologies

1.2 Importance in Modern Organizations

The contemporary organizations are very connected, and in most cases, they cut across several geographies using cloud and hybrid networks. Security technologies are essential to safeguard sensitive business data and ensure that the regulatory frameworks like GDPR and ISO 27001 are complied with and that the trust of the customers is not lost. Cyber incidents, be it financial or reputational, have been costly over the past few years, making the need to actively implement security technologies paramount (Kidd, 2024).



Figure 2 : Importance of Information Security in Organization

1.3 Aim and Objectives of the Report

The purpose of the report is to have a full understanding of security technologies, evaluate their effectiveness and the use of these technologies in real life situations based on case studies. The objectives include:

- Assessing the most important forms of security technologies in institutions.
- Researching issues and barriers to implementation.
- Evaluating lessons learned arising out of recent cyber incidents.
- Proposing the best practices towards enhanced security management.

1.4 Scope and Limitations

The report is dedicated to the organizational security technologies adopted between 2020-2024, and the concentration is made on the network, cloud, data security, and new technologies. Although it is based on publicly available case studies, certain proprietary technical information might not be available because of confusion.

2. Rationale and Objectives

2.1 Need for Security Technology

Organizations are becoming more dependent on digital infrastructures, which expose them to advanced cyber threats. Security technologies offer measures to curb illegal

access, identify malicious actions, and react effectively to the incidents. The fast incorporation of cloud computing, remote working, and the IoT gadgets has also increased the demand to put into place high-level security practices (simenove, 2024).

2.2 Goals of Security Technologies Implementation



Figure 3 : Objectives of Security Technologies

- Secure organizational and confidential customer information.
- Make sure that it is socially of responsible.
- Losses in operations through cyber incidences.
- Monitoring tools and automation to identify and react to threats on a real-time basis.
- Create an environment of security consciousness among the employees.

2.3 Research Questions

- Which are the basic security technologies implemented in organizations?

- To what extent do these technologies reduce contemporary cyber threats?
- What can be taught as a result of high-profile security breaches in the recent past?
- What can organizations do to enhance security technologies implementation strategies? If read more all answer this question then click on [Appendix A](#).

3. Literature Review

3.1 concept of information security

The protection of an organization from unauthorized access or loss by an unqualified third party. Information security (InfoSec) is the art of the safeguarding information, to make sure that it does not fall into the wrong hands, is not revealed, altered, or destroyed. In any contemporary organization, data is among the most precious resources, which may be customer information, financial information, intellectual information, and operational information.

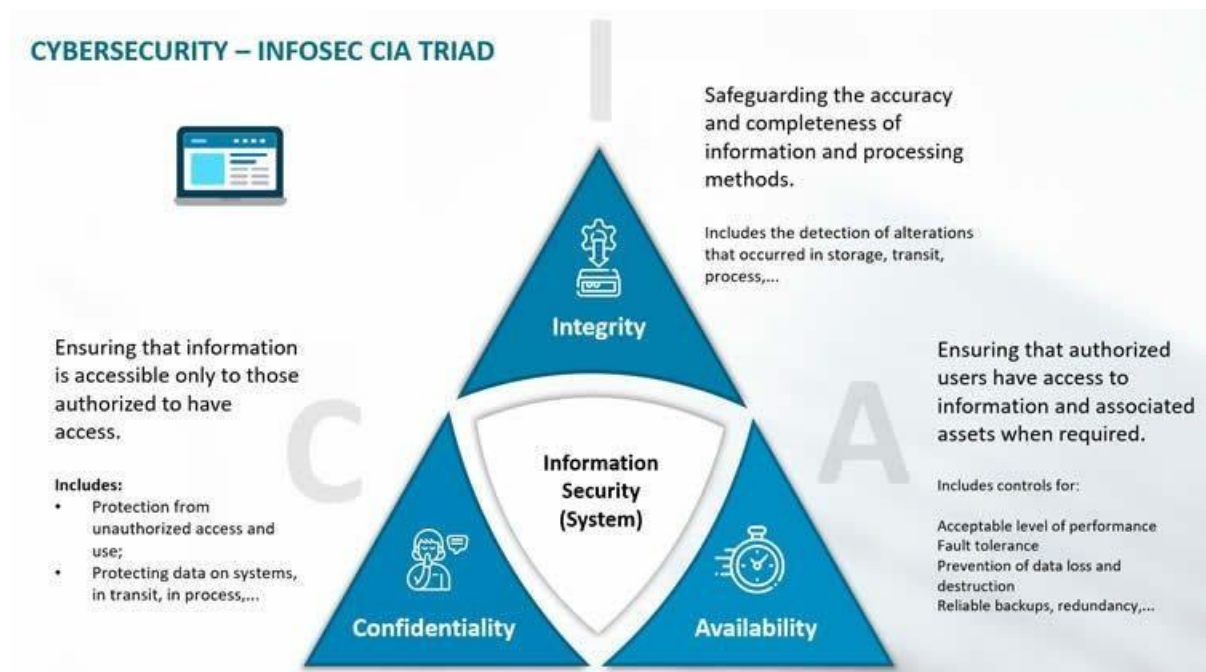


Figure 4 : Information Security Concept and CIA Triad

This data should be ensured to protect the trust of a client and stakeholders and to guarantee continuity of the business not to lose financial or reputational losses. The CIA triad of confidentiality, integrity, and availability are used to guide information security. Sensitive information can only be accessed by authorized personnel due to confidentiality, data can only be accurate and free of any alterations due to integrity,

and can be made available at any time necessary due to availability. To limit liability and to safeguard data at all stages of its life cycle in the ever-digital and interconnected world, organizations deploy multi-layered security measures, integrating hardware, software, and human processes, including firewalls, encryption, antivirus software, access controls, employee training and incident response plans.

3.2 Security Technology types



Figure 5 : Types of Security

- **Technologies Network:** Security Technologies Firewalls, intrusion detection/prevention system (IDS/IPS), VPNs, network segmentation.
- **Endpoint Security:** Antivirus, anti-malware, endpoint detection and response (EDR) solutions.
- **Application Security:** Secure coding instructions, scanning vulnerability, web application firewall.
- **Cloud Protection:** IAM, encryption of data, round-the-clock surveillance.

3.3 Standards and Frameworks

- **ISO 27001:** Gives a systematic manner of management of information security.
- **NIST Cybersecurity Framework:** Helps organizations to detect, respond, and recover, and identify and protect against cyber threats.
- **MITRE ATT&CK:** Adversary threat modelling tactics and techniques knowledge base.

3.4 Recent Trends of Security Technologies

- **Artificial Intelligence-Based security:** Threat identification, anomaly detection, and predictive analytics.
- **Blockchain:** Improving data integrity and transaction security.
- **Zero-Trust Architecture:** Does not trust anybody and authentication is necessary.
- **SIEM and Cloud Platforms Integration:** Real time monitoring and responding to an incident.

4. Core Security Technologies

Network security: technologies are needed to secure organizational networks against infiltrations and attacks. Firewalls are used as the initial defense mechanism that prevents bad traffic and permits the normal communication. Intrusion Detection and Prevention Systems (IDS/IPS) are used as systems that monitor network traffic to identify and prevent suspicious network traffic. VPNs are also used to gain secure remote access and network segmentation prevents the threat travel laterally through the network.

Advanced Cybersecurity Solutions

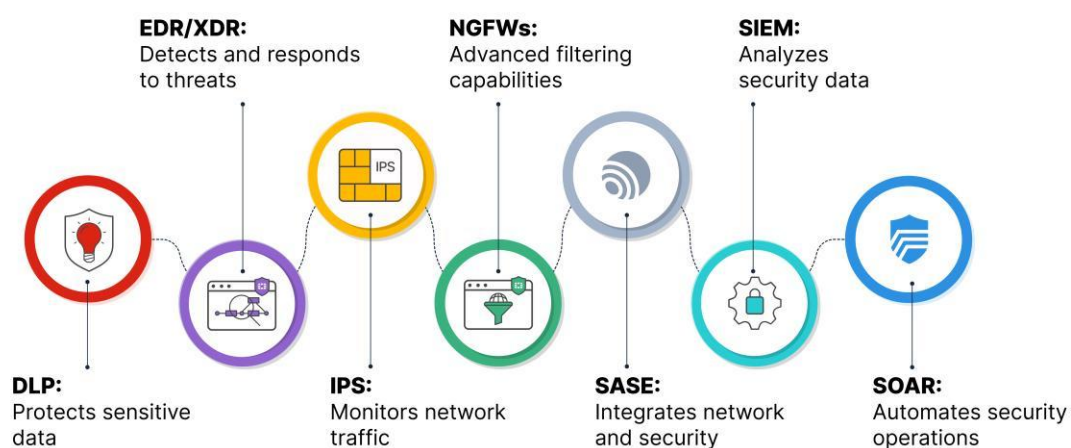


Figure 6 : Technologies for the Modern Enterprise

Data protection : technologies are aimed at protecting the sensitive information. Encryption will make sure that the data cannot be read by unauthorized parties, both in storage and transmission. The access controls ensure authentication and authorization such that only legitimate users can have access to specific system or files. Data Loss Prevention (DLP) systems upkeep and stop sensitive information leakage or mishandling.

4 Common Cyber Threats

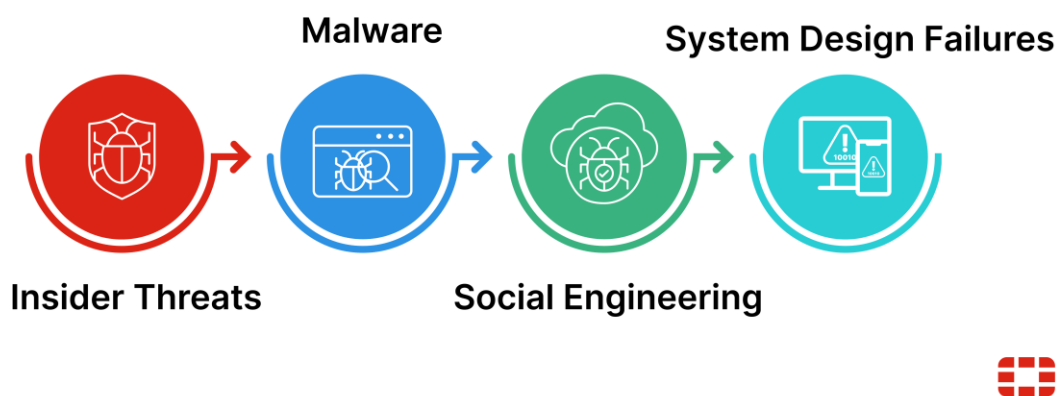


Figure 7: Cyber Threat

The physical and biometric security: measures are used to safeguard the organization and its premises. Physical areas are monitored by CCTV cameras and surveillance, and biometric authentication (fingerprint or facial recognition) enhances the security of access. Smart cards and security tokens offer another layer of security due to multi-factor authentication, which will minimize the risk of entry without permission.

Emerging technologies: There are new security technologies which are meant to deal with complex threats. The structures powered by AI can identify suspicious patterns and forecast the possible attacks prior to their occurrence. With blockchain technology, the integrity of data and tracing transactions is guaranteed and therefore

tampering becomes more challenging. The zero-trust model verifies users and devices on a constant basis, which presupposes that no other party is deemed to be a trustworthy one and enhances the general security posture.



Figure 8 : New Technology

5. Security Technology and their Applications

The technology of security is used within organizations to secure the networks, data and systems against cyber attacks. Firewalls and IDS/IPA provide security to networks by blocking intruders and identifying malicious actions. Encryption, and cryptography ensure that sensitive information is not read or modified by the unauthorized parties. Access control and authentication systems restrict access to systems and resources to authorized staff members. Malware and antivirus protection identify, block, and delete malicious software that may cause havoc to the operations or steal data. Security Information and Event Management (SIEM) software offers centralized control of security incident and real-time warning enabling quicker identification and reaction to the possible danger. The combination of these technologies forms a strong environment of security minimizing risks and guaranteeing continuity of the business.

6. Case Studies on Security Technologies

6.1 Case Study 1: Ransomware Nights against Royal Mail (UK)

In 2023, Royal Mail, a part of the largest post and logistics organization in the United Kingdom, was struck with a major ransomware attack, which disrupted its operations considerably. The cyber attack entailed the use of malicious software which encrypted some of the critical internal systems hence rendering the employees unable to access necessary files, applications, and working tools. This did not only stop the daily operations but also created a great risk to the customer service, delivery schedule, and sensitive business details. It was found that the endpoint protection systems of the organization could not identify the ransomware, which indicates the weakness of the software and system monitoring guidelines. Moreover, network segmentation was not satisfactory and this enabled the ransomware to spread through several systems with ease (Street, 2023).

Royal Mail acted by installing the backups to recover failed systems and isolating the damaged machines to avoid additional damages. Although these measures had contributed in decreasing the operational downtime, the incident has revealed the vulnerability in the overall security strategy of the organization. The case pointed out that organizations should not rely on technical defense alone they should adopt a multi-layered system of defence that includes endpoint security, strong network controls, and human-centered solutions such as frequent patching and training of employees on cybersecurity. One of the other lessons of the attack was the significance of well tested incident response plans in case of ransomware breach to ensure that the response is prompt and efficient (Danielson, 2025).

6.2 Case study 2 : Cloud breach at Microsoft azure (Storm-0558 follow-up)

In 2024, another major cloud security incident occurred in Microsoft Azure, whereby its cloud resources were accessed unauthorized. Access controls were misconfigured, which resulted in the breach where attackers used vulnerabilities in Identity and Access Management (IAM) policies. Also, surveillance systems could not warn the administrators in time and postponed the process of detecting suspicious activity. This incident showed some severe weaknesses in cloud infrastructure management as it was evident that even sophisticated cloud systems can be prone to misconfiguration and management (farris, 2024).

The answer to these events provided by Microsoft is that it implemented multi-factor authentication (MFA), better real-time user activity monitoring, and more vigorous practices on cloud security posture management. The attack acted as an alert that cloud environments need constant monitoring since the misconfiguration can easily turn into a critical security vulnerability. The major information gained during this event is that it is necessary to perform configuration audits on a regular basis, provide sophisticated monitoring and alerting mechanisms, and implement a zero-trust security model, i.e., in which no user or device receives a default trust. This will mean that in case the access controls have not been properly set, suspicious behavior can be identified and counteracted before it can cause significant loss of data or exposure of operations. (Archer, 2025)

7. Issue Identification and Challenges

Various technical factors that organizations encounter in the maintenance of security include the use of outdated legacy systems, intricate cloud and hybrid platforms, and exponentially changing cyber threats. There are human and organizational risks like insufficiency of employee training, being a victim of social engineering attack, and unwillingness to use new security technologies, among others. Moreover, legal and ethical issues also demand that organizations should adhere to such regulations as GDPR and HIPAA but should be responsible when working with sensitive data. The security in the emerging threats such as AI-powered malware, ransomware-as-a-service and supply-chain attacks also complicate security management. The solution to these problems involves technical solutions and active organizational approaches.

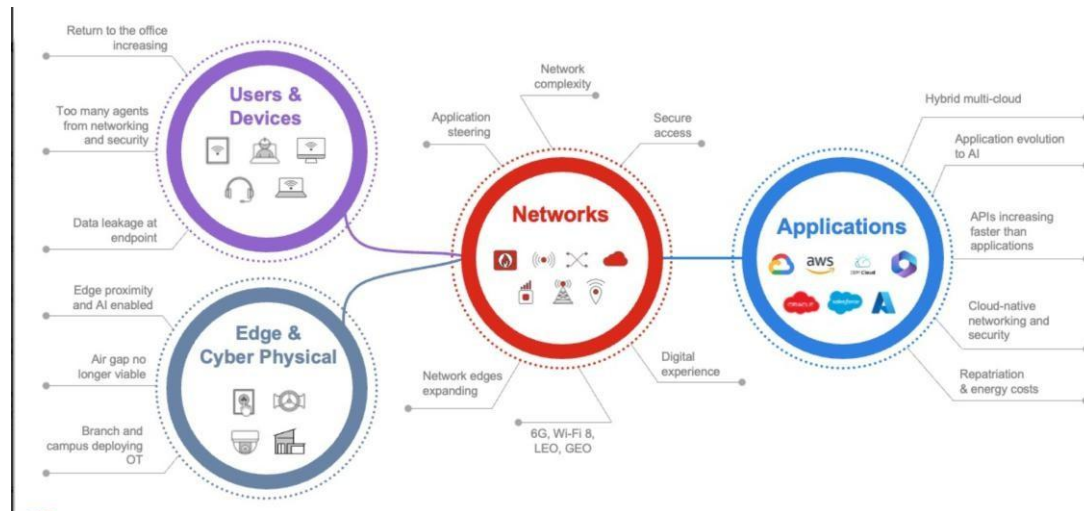


Figure 9 : Security Challenges

8. Best Practices in Implementation

In order to achieve the security, organizations ought to have well defined policies and governance structures in respect to security. The frequent risk assessment and vulnerability scanning procedures allow the deficiencies to be detected prior to their exploitation. The training and employee awareness programs make employees aware of security procedures and know how to identify threat like phishing. Incident response and constant monitoring enable an organization to respond and identify attackers promptly. The implementation of emerging technologies, such as AI-based threat detection and zero-trust architectures, can increase the overall security posture and make organizations resistant to impingements of the changed cyber threat.

9. Conclusion

In the present organization, security technologies are essential to safeguard the digital resources and ensure the continuity of operation. As the evaluation of Royal Mail and

Microsoft Azure breaches shows, even the most sophisticated security technologies cannot be efficient when they are not sufficiently configured, maintained, and linked to human operations. The multi-layered approach of network, data, cloud, and physical security, coupled with training and risk management strategies of the employees, should be adopted by organizations. New technologies like AI and blockchain provide new protection and predictive features. The best practices, continuous monitoring systems, and learning of the past incidences can help organizations improve their security posture and mitigate exposure to new cyber threats.

10. References

- Archer, D. (2025, may 21). *level blue*. Retrieved from Storm-0558 and the Dangers of Cross-Tenant Token Forgery: <https://levelblue.com/blogs/spiderlabs-blog/storm-0558-and-the-dangers-of-cross-tenant-token-forgery>
- Danielson, L. (2025, dec 2). *Huntress* . Retrieved from Royal Mail Ransomware Attack: <https://www.huntress.com/threat-library/ransomware/royal-mail-ransomware>
- farris, c. (2024, 04 13). *Microsoft (Storm-0558)*. Retrieved from Public Cloud Security Breaches: <https://www.breaches.cloud/incidents/o365-2023/>
- Kidd, C. (2024, oct 09). *Splunk*. Retrieved from https://www.splunk.com/en_us/blog/learn/cybersecurity-frameworks.html
- simenove, m. (2024, sep 04). *amatas*. Retrieved from <https://amatas.com/blog/cybersecurity-challenges-in-2024-key-issues-and-solutions/>
- Street, S. (2023, march 7). *Cyber Management Alliance* . Retrieved from Royal Mail Ransomware Attack Timeline: <https://www.cm-alliance.com/cybersecurity-blog/royal-mail-ransomware-attack-timeline>
- Valorem Reply. (2025, May 16). Retrieved from <https://www.valoremreply.com/resources/insights/blog/types-of-cybersecurity>

11. Appendix

Appendix A

1. What are the fundamental security technology used in organizations?

There are various security technologies used by organizations to safeguard information, networks and users. The rudimentary technologies are as follows:

a) Network Security Technologies.

- Firewalls to regulate the inbound and outbound traffic.
- Detection and Prevention Systems (DDoS)
- Virtual Privatization Networks (VPN)
- Secure gateways, network segmentation.

b) Endpoint Security

- Antivirus/ Anti-malware software.
- Endpoint Detection and Response (EDR).
- Device administration and patch administration.

c) Technologies of Data Protection.

- Data rest and data transmission encryption.
- Data Loss Prevention (DLP)
- Backup and recovery systems

d) Identity and Access Management.

- Below is a description of multi-factor authentication (MFA).
- Role-based access control (RBAC)
- Privileged access control

e) Monitoring and Response

- Security Information and Event Management (SIEM).
- Security Operations Centers (SOC)
- Incident response platforms are websites dedicated to responding to incidents
- Incident response platforms Incident response platforms are web sites which focus on responding to incidents.
- These technologies provide a layered defense (defense-in-depth) in order to protect organizational assets.

2. How much of these technologies minimize modern cyber threats?

- The security technologies can lower the cyber threats by a substantial margin, yet they are not able to cut down the risk altogether.

Prevention:

- Firewalls, MFA, and encryption prevent unauthorized access to sensitive data and block it.

Detection:

- The unfamiliar behavior identified by SIEM and IDS/IPS can be ransomware or phishing.

Response:

- Incident response tools contribute to the isolation of affected systems as well as the restoration of services in the organization in a shorter period.

Effectiveness:

- Research indicates that MFA is able to forestall over 90 percent of account takeover attacks.
- Encryption secures information even in case of system intrusion.
- The constant monitoring makes the detection time shorter by months to hours.

Limitations:

- When the tools are misconfigured (e.g. cloud IAM errors).
- Human mistake and social trickery.
- Outside controls on zero-day vulnerabilities.
- Thus, technologies can only be effective in case of combining it with the policies, training, and risk management.

3. What can be learned following high profile security breaches in the recent past?

The lessons can be learned through such recent incidents as the Royal Mail ransomware (2023) or Microsoft Azure Storm-0558 (2024):

Technology in Itself is not Adequate.

- Advanced systems also collapsed because of improper configuration and monitoring.

Patch Management Significance.

- Late disclosure of vulnerabilities gave attackers a chance to exploit known vulnerabilities.
- Need for Zero-Trust Approach
- Reliable users or networks are relied upon and add more harm.
- Human Factor Is Critical

Technical controls are avoided by phishing and social engineering.

- Incident Response Preparedness Incident response preparedness (IRP) involves essential preparations and actions designated to facilitate and synchronize response operations during a cyber attack or incident
- Incident Response Preparedness Incident response preparedness (IRP) is a collection of necessary preparations and activities that are designed to streamline and coordinate response efforts in response to a cyber attack or incident.
- Companies that had backup and response strategies went back to business more quickly.
- Cloud Security Responsibility.
- The responsibility of cloud breaches is shared by the provider and the customer.
- These violations are lessons that security should be an ongoing process, multifaceted, and organization-wide.

4. How can organizations improve the use of security technologies?

In order to enhance performance, organizations must:

a) Strengthen Governance

Formulate security policies that are consistent with ISO 27001 and NIST.

Delimitate roles and responsibilities.

b) Make Technical Controls Improved.

- Introduce MFA, encryption and SIEM.
- Routine vulnerability scanning and fixing.
- Eliminate-zero-trust architecture.

c) Enhance Human Awareness

- Security awareness awareness training.

Phishing simulations

Good culture of incident reporting.

d) Continuous Monitoring

24/7 SOC operations

Intelligence integration of threats.

Regular security audits

e) Incident Preparedness

Disaster recovery plans and backup plans.

Frequent penetration test.

The lessons-learned post-incident reviews.

Organizations can also achieve the maximum benefit of security technologies by combining people, process, and technology.